

Data Structures and Algorithms

Question 1

Given a sorted array, which technique can find whether there exists a triplet with sum = X in $O(n^2)$ time?

- (A) Nested Loops
- (B) Binary Search only
- (C) Two Pointer Technique
- (D) DFS

Ans: C

Question 2

What is the minimum number of swaps required to reverse an array of size n in-place?

- (A) n
- (B) $n - 1$
- (C) $\lfloor n/2 \rfloor$
- (D) $\lceil n/2 \rceil$

Ans: C

Question 3

Which of the following operations in an array requires shifting of elements?

- (A) Accessing an element
- (B) Updating an element
- (C) Insertion at a specific position
- (D) Binary Search

Ans: C

Question 4

In a circular array of size n, the next index after i is calculated using:

- (A) $(i+1)$
- (B) $(i + 1) \% n$
- (C) $(i - 1 + n) \% n$
- (D) $i * n$

Ans: B

Question 5

Elements 1, 2, 3, 4, 5 are inserted in a stack in sequence. The elements are printed while popping them out. Which of the following is wrong print sequence?

- (A) 3 5 4 2 1

- (B) 1 3 2 4 5
 - (C) 5 4 3 2 1
 - (D) 2 4 1 5 3
- Ans. D

Question 6

Which of the following postfix expressions is equivalent to the infix expression $(A + B) * (C - D)$?

- (A) $AB+CD-*$
- (B) $ABCD+-*$
- (C) $AB+CD-.*+$
- (D) $ABCD-.*+$

Ans: A

Question 7

Which of the following is TRUE about infix, prefix, and postfix expressions?

- (A) Postfix expressions require parentheses
- (B) Prefix expressions require operator precedence rules
- (C) Postfix expressions can be evaluated using a stack
- (D) Infix expressions do not use operators

Ans: C

Question 8

Consider the postfix expression:

$5\ 6\ 2\ +\ *\ 12\ 4\ /\ -$

The result of the expression is:

- (A) 37
- (B) 40
- (C) 43
- (D) 45

Ans: A

Question 9

Suppose the postorder traversal of a BST is:

$2\ 4\ 3\ 8\ 7\ 5$

The root of the BST is:

- (A) 3
- (B) 5

- (C) 7
- (D) 8

Ans: B

Question 10

Consider the following sequence of operations on an empty stack:

Push(5), Push(8), Push(2), Pop(), Push(9), Pop()

Which element remains at the top of the stack?

- (A) 5
- (B) 8
- (C) 2
- (D) 9

Ans: B

Question 11

What is the time complexity of PUSH and POP operations in a stack implemented using arrays?

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n \log n)$

Ans: C

Question 12

In an array implementation of stack, stack overflow occurs when:

- (A) Stack becomes empty
- (B) $TOP = 0$
- (C) $TOP = MAX - 1$
- (D) $TOP = -1$

Ans: C

Question 13

If a stack is implemented using a linked list, overflow occurs when:

- (A) $TOP = NULL$
- (B) Stack size reaches fixed limit
- (C) System memory is exhausted
- (D) Last node points to first node

Ans: C

Question 14

A circular queue of size n is implemented using an array. The queue is considered full when:

- (A) $\text{FRONT} == \text{REAR}$
- (B) $(\text{REAR} + 1) \% n == \text{FRONT}$
- (C) $(\text{FRONT} + 1) \% n == \text{REAR}$
- (D) $\text{REAR} == n - 1$

Ans: B

Question 15

Consider the following queue operations performed on an initially empty queue:

Enqueue(5), Enqueue(10), Dequeue(), Enqueue(15), Enqueue(20), Dequeue()

The final front element of the queue is:

- (A) 5
- (B) 10
- (C) 15
- (D) 20

Ans: C

Question 16

A queue is implemented using two stacks $S1$ and $S2$. Insert operation pushes elements into $S1$, while delete operation transfers elements from $S1$ to $S2$ whenever $S2$ is empty.

What is the amortized time complexity of the delete operation?

- (A) $O(1)$
- (B) $O(\log n)$
- (C). $O(n)$
- (D). $O(n \log n)$

Ans: C

Question 17

In a priority queue implemented using a max-heap containing n elements, insertion of a new element takes:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

Ans: B

Question 18

A binary tree has exactly 20 nodes with degree 2. The number of leaf nodes in the tree is:

- (A) 19
- (B) 20
- (C) 21
- (D) 22

Ans: C

Question 19

Consider the following operations on an empty circular queue of size 5:

Enqueue(1), Enqueue(2), Enqueue(3), Dequeue(), Enqueue(4), Enqueue(5), Enqueue(6)

What happens during insertion of 6?

- (A) Queue overflow
- (B) Queue underflow
- (C) Insertion succeeds normally
- (D) Front element gets deleted automatically

Ans: C

Question 20

A queue contains the following elements from front to rear:

2, 4, 6, 8

After performing two dequeue operations and one enqueue operation of 10, the queue becomes:

- (A) 6, 8, 10
- (B) 4, 6, 8, 10
- (C) 2, 4, 10
- (D) 8, 10

Ans: A

Question 21

A queue implemented using two stacks has worst-case deletion complexity:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

Ans: C

Question 22

In a linked-list implementation of queue, which pointer is necessary to perform both enqueue and dequeue in $O(1)$ time?

- (A) Only front pointer
- (B) Only rear pointer
- (C) Both front and rear pointers
- (D) No pointer required

Ans: C

Question 23

Consider two sorted arrays which are having sizes P and Q. If merge algorithm is used to combine those arrays into a single array then total comparisons and movements required respectively are

- (A) $O(P^2)$ and $O(P + Q)$
- (B) $O(P)$ and $O(P + Q)$
- (C) $O(P + Q)$ and $O(Q)$
- (D) $O(P + Q)$ and $O(P + Q)$

Ans: D

Question 24

What is the minimum number of shifts required to insert an element at the beginning of an array of size n?

- (A) 1
- (B) $n/2$
- (C) n
- (D) $n - 1$

Ans: C

Question 25

The preorder and inorder traversals of a binary tree are:

Preorder: A B D E C F

Inorder: D B E A F C

The postorder traversal is:

- (A) D E B F C A
- (B) E D B F C A
- (C) D E F B C A
- (D) E D F C B A

Ans: A

Question 26

A singly linked list has n nodes. Which of the following operations can be performed in $\theta(1)$ time in the worst case?

- (A) Deleting the last node
- (B) Searching an element
- (C) Inserting a node at the beginning
- (D) Finding the middle node

Ans: C

Question 27

Consider the following code segment:

```
Node *p = head;
```

```
while(p != NULL && p->next != NULL)
{
    p = p->next->next;
}
```

If the linked list contains 9 nodes, then after execution, p points to:

- (A) 7th node
- (B) 8th node
- (C) 9th node
- (D) NULL

Ans: C

Question 28

The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19. Which one of the following is the postorder traversal of the tree?

- (A) 20, 19, 18, 16, 15, 12, 11, 10
- (B) 11, 12, 10, 16, 19, 18, 20, 15
- (C) 10, 11, 12, 15, 16, 18, 19, 20
- (D) 19, 16, 18, 20, 11, 12, 10, 15

Ans: B

Question 29

Consider the following statements regarding linked lists:

1. Linked lists provide dynamic memory allocation.
2. Random access is efficient in linked lists.
3. Insertion/deletion in the middle is efficient if pointer is given.

Which of the following is correct?

- (A) 1 and 2 only
- (B) 2 and 3 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

Ans: C

Question 30

Which of the following statements about Binary Search is FALSE?

- (A) It requires sorted data
- (B) It can be implemented recursively
- (C) Its worst-case complexity is $\Theta(\log n)$
- (D) It works efficiently on singly linked lists

Ans: D

Question 31

A doubly linked list contains n nodes. Given a pointer to a node p , deletion of p requires updating:

- (A) One pointer only
- (B) Two pointers only
- (C) Three pointers only
- (D) Four pointers only

Ans: B

Question 32

Consider the following code:

```
Node *p = head;
int cnt = 0;

while(p != NULL)
{
    cnt++;
    p = p->next;
}
```

If the linked list contains 15 nodes, the value of “cnt” after execution will be:

- (A) 14
- (B) 15
- (C) 16
- (D) Undefined

Ans: B

Question 33

Consider the following function:

```
Node* fun(Node *head)
{
    if(head == NULL || head->next == NULL)
        return head;

    Node *p = head->next;
```



```

    head->next = p->next;
    p->next = head;

    return p;
}

```

If the input linked list is:

10 → 20 → 30 → 40

What will be the modified list?

- (A) 20 → 10 → 30 → 40
- (B) 40 → 30 → 20 → 10
- (C) 10 → 30 → 20 → 40
- (D) 20 → 30 → 40 → 10

Ans: A

Question 34

In a Binary Search Tree, the search operation takes $O(h)$ time, where 'h' is the height of the tree. In the worst case, the height of a BST containing n nodes becomes:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

Ans: C

Question 35

A circular queue implemented using an array of size n initially has:

FRONT = 2 and REAR = 5

After one insertion operation, the new value of REAR becomes:

- (A) $(\text{REAR} - 1) \% n$
- (B) $(\text{REAR} + 1) \% n$
- (C) $(\text{FRONT} + 1) \% n$
- (D) $\text{REAR} + \text{FRONT}$

Ans: B

Question 36

Consider a binary max-heap implemented using an array. Which one of the following array represents a binary max-heap?

- (A) 25, 12, 16, 13, 10, 8, 14

- (B) 25, 12, 16, 17, 10, 8, 14
- (C) 25, 14, 16, 13, 10, 8, 12
- (D) 25, 14, 12, 13, 10, 8, 16

Ans: C

Question 37

The minimum number of nodes in a binary tree having a total of 5 levels is:

- (A) 32
- (B) 5
- (C) 63
- (D) 9

Ans: B

Question 38

The inorder traversal of tree will yield a sorted listing of elements of tree in

- (A) Binary trees
- (B) Binary search trees
- (C) Heaps
- (D) None of above

Ans: B

Question 39

A full binary tree with N non-leaf nodes contains a total of:

- (A) $\log N$ nodes
- (B) $N+1$ nodes
- (C) $2N$ nodes
- (D) $2N+1$ nodes

Ans: D

Question 40

The following sequence is inserted into a binary search tree:

6 10 25 12 4 7 15 8 33

What is the type of traversal given by?

6 4 10 7 25 8 12 33 15

- (A) Inorder
- (B) Postorder
- (C) Preorder
- (D) Level order

Ans: D

Question 41

Which sorting algorithm is stable and has worst-case time complexity $O(n \log n)$?

- (A) Quick sort
- (B) Heap sort
- (C) Merge sort
- (D) Selection sort

Ans: C

Question 42

In Quick sort, the worst-case occurs when:

- (A) Pivot is median
- (B) Pivot divides array equally
- (C) Pivot is always smallest or largest
- (D) Array is random

Ans. C

Question 43

The recurrence relation for binary search is:

- (A) $T(n) = T(n-1) + c$
- (B) $T(n) = 2T(n/2) + c$
- (C) $T(n) = T(n/2) + c$
- (D) $T(n) = T(n) + c$

Ans. C

Question 44

What is recurrence for worst case of Quick Sort and what is the time complexity in Worst case?

- (A) $T(n) = T(n-2) + O(n)$ and $O(n^2)$
- (B) $T(n) = T(n-1) + O(n)$ and $O(n^2)$
- (C) $T(n) = 2T(n/2) + O(n)$ and $O(n \log n)$
- (D) $T(n) = T(n/10) + T(9n/10) + O(n)$ and $O(n \log n)$

Ans: B

Question 45

A BST contains the elements:

15, 10, 20, 8, 12, 17, 25

Which node is the inorder successor of 15?

- (A) 10
- (B) 12
- (C) 17
- (D) 20

Ans: C

Question 46

Which of the following sequences cannot represent preorder traversal of a BST?

- (A) 40, 30, 35, 80, 100
- (B) 40, 30, 35, 20, 80
- (C) 8, 5, 1, 7, 10, 12
- (D) 30, 20, 10, 15, 25, 40, 50

Ans: B

Question 47

Which of the following arrays represents a valid max-heap?

- (A) 50, 40, 45, 30, 35, 20, 10
- (B) 50, 60, 45, 30, 35, 20, 10
- (C) 40, 50, 45, 30, 35, 20, 10
- (D) 50, 40, 45, 60, 35, 20, 10

Ans: A

Question 48

What is the maximum number of nodes possible in a binary heap of height h ?

- (A) 2^h
- (B) $2^h - 1$
- (C) $2^{h+1} - 1$
- (D) 2^{h+1}

Ans: C

Question 49

Consider the following algorithm:

```
for(i = 1; i <= n; i++)
{
    for(j = i; j <= n; j = j * 2)
    {
        // constant time operation
    }
}
```

What is the time complexity of the algorithm?

- (A) $\Theta(n)$
- (B) $\Theta(n \log n)$
- (C) $\Theta((\log n)^2)$
- (D) $\Theta(n^2)$

Ans: B

Question 50

Linear Search on an array of size n has average-case complexity $\Theta(n)$ because, on average, the searched element is expected to be found after examining

- (A) One element
- (B) $\log n$ elements
- (C) $n/2$ elements
- (D) n^2 elements

Ans: C